

Homework_2 - Solutions

Math 742

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3.1 Find expressions for $\mathbb{E}(X)$ and $\mathbb{V}(X)$ if X is a random variable with the general discrete uniform(N_0, N_1) distribution that puts equal probability on each of the values $N_0, N_0+1, \dots, N_1$. Where $N_0 \leq N_1$ and N_0 and N_1 are both integers.

$$\mathbb{P}(X = k) = \frac{1}{m}; k = N_0, \dots, N_1; m = N_1 - N_0 + 1$$

$$\begin{aligned}\mathbb{E}(X) &= \sum_{i=0}^{m-1} (N_0 + i) \frac{1}{m} = \frac{1}{m} \sum_{i=0}^{m-1} (N_0 + i) \\ &= \frac{1}{m} \left(\sum_{i=0}^{m-1} N_0 + \sum_{i=0}^{m-1} i \right) = \frac{1}{m} \left(mN_0 + \frac{(m-1)m}{2} \right) = N_0 + \frac{m-1}{2}\end{aligned}$$

Now plug in $N_1 - N_0 + 1$ for m ,

$$= N_0 + \frac{N_1 - N_0 + 1 - 1}{2} = \frac{2N_0 + N_1 - N_0}{2} = \frac{N_0 + N_1}{2}$$

Or, the easy way is to recognize that for a uniform distribution the mean is the point midway between the left endpoint and the right endpoint.

To find the variance, shift all values of X by N_0 .

$$Y = X - N_0$$

Now, $\mathbb{V}(X) = \mathbb{V}(Y)$.

Y is uniform on the interval $[0, m-1]$.

$$\mathbb{E}(Y) = \frac{m-1}{2}, \mathbb{E}(Y^2) = \frac{1}{m} \sum_{i=0}^{m-1} i^2 = \frac{1}{m} \frac{m(m-1)(2m-1)}{6} = \frac{(m-1)(2m-1)}{6}$$

$$\mathbb{V}(Y) = \mathbb{E}(Y^2) - [\mathbb{E}(Y)]^2 = \frac{(m-1)(2m-1)}{6} - \left(\frac{m-1}{2} \right)^2 = \frac{m^2-1}{12}$$

In terms of N_0 and N_1 (i.e. $m = N_1 - N_0 + 1$)

$$\mathbb{E}(X) = \frac{N_0 + N_1}{2} \text{ and } \mathbb{V}(X) = \mathbb{V}(Y) = \frac{(N_1 - N_0 + 1)^2 - 1}{12} = \frac{(N_1 - N_0)(N_1 - N_0 + 2)}{12}$$

3.4 A man with n keys wants to open his door and tries the keys at random. Exactly one key will open the door. Find the mean number of trials if,

(a) unsuccessful keys are not eliminated from further selections.

i.e. with replacement

Each trial is an independent random pick from the n keys, so the success probability each time is $p = \frac{1}{n}$

This is a geometric distribution since we are looking for the trial where the first success occurred.

$$\mathbb{E}(X) = \frac{1}{p} = n$$

(b) unsuccessful key are eliminated.

i.e. without replacement

Now he is effectively trying keys in a random order (a random permutation of the n keys).

The correct key is equally likely to appear in any of the n positions, so

$$\mathbb{P}(X = k) = \frac{1}{n}, \text{ where } k = 1, 2, \dots, n$$

So this is a discrete uniform distribution on $\{1, 2, \dots, n\}$

$$\mathbb{E}(X) = \frac{n+1}{2}$$

3.23 The *Pareto Distribution*, with parameters α and β , has pdf

$$f(x) = \frac{\beta \alpha^\beta}{x^{\beta+1}}, \text{ where } \alpha < x < \infty, \alpha > 0, \beta > 0$$

(a) Verify that $f(x)$ is a PDF.

- First, $f(x) \geq 0$ for all $x > \alpha$ since all parameters are positive.
- Now check that the total probability is 1:

$$\int_{\alpha}^{\infty} \frac{\beta \alpha^\beta}{x^{\beta+1}} dx = \beta \alpha^\beta \int_{\alpha}^{\infty} x^{-(\beta+1)} dx.$$

Since

$$\int x^{-(\beta+1)} dx = \frac{x^{-\beta}}{-\beta},$$

we obtain

$$\beta \alpha^\beta \left[\frac{x^{-\beta}}{-\beta} \right]_{\alpha}^{\infty} = \beta \alpha^\beta \left(0 - \frac{\alpha^{-\beta}}{-\beta} \right) = 1.$$

Therefore $f(x)$ is a valid probability density function.

(b) Derive the mean and variance of this distribution

The mean is

$$\mathbb{E}[X] = \int_{\alpha}^{\infty} x f(x) dx = \int_{\alpha}^{\infty} x \frac{\beta \alpha^{\beta}}{x^{\beta+1}} dx = \beta \alpha^{\beta} \int_{\alpha}^{\infty} x^{-\beta} dx.$$

For $\beta \leq 1$, the integral diverges. For $\beta > 1$,

$$\int x^{-\beta} dx = \frac{x^{1-\beta}}{1-\beta},$$

so,

$$\mathbb{E}[X] = \beta \alpha^{\beta} \left[\frac{x^{1-\beta}}{1-\beta} \right]_{\alpha}^{\infty} = \beta \alpha^{\beta} \left(0 - \frac{\alpha^{1-\beta}}{1-\beta} \right) = \frac{\beta \alpha}{\beta - 1}.$$

Thus,

$$\mathbb{E}[X] = \frac{\beta \alpha}{\beta - 1}, \quad \beta > 1.$$

Next, compute the second moment:

$$\mathbb{E}[X^2] = \int_{\alpha}^{\infty} x^2 f(x) dx = \beta \alpha^{\beta} \int_{\alpha}^{\infty} x^{1-\beta} dx.$$

For $\beta > 2$,

$$\int x^{1-\beta} dx = \frac{x^{2-\beta}}{2-\beta},$$

and hence

$$\mathbb{E}[X^2] = \beta \alpha^{\beta} \left[\frac{x^{2-\beta}}{2-\beta} \right]_{\alpha}^{\infty} = \frac{\beta \alpha^2}{\beta - 2}.$$

Therefore, for $\beta > 2$,

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{\beta \alpha^2}{\beta - 2} - \left(\frac{\beta \alpha}{\beta - 1} \right)^2 = \frac{\beta \alpha^2}{(\beta - 1)^2 (\beta - 2)}.$$

(c) Prove that the variance does not exist if $\beta \leq 2$.

The variance exists only if $\mathbb{E}[X^2] < \infty$. From above,

$$\mathbb{E}[X^2] = \beta \alpha^{\beta} \int_{\alpha}^{\infty} x^{1-\beta} dx.$$

If $\beta < 2$, then $1 - \beta > -1$, and the improper integral diverges:

$$\int_{\alpha}^{\infty} x^{1-\beta} dx = \infty.$$

If $\beta = 2$, then

$$\int_{\alpha}^{\infty} x^{-1} dx = [\ln x]_{\alpha}^{\infty} = \infty.$$

Thus $\mathbb{E}[X^2] = \infty$ for all $\beta \leq 2$, and so the variance does not exist.

$$\boxed{\text{Var}(X) \text{ exists if and only if } \beta > 2.}$$

6.2 Let $X_1, X_2, \dots, X_n$ be independent random variables with densities

$$f_{X_i}(x|\theta) = \begin{cases} e^{i\theta-x} & : x \geq i\theta \\ 0 & : x < i\theta \end{cases}$$

Prove that $T = \min(X_i/i)$ is a sufficient statistic for θ .

Sufficiency of $T = \min_i(X_i/i)$

Let $X_1, \dots, X_n$ be independent with densities

$$f_{X_i}(x | \theta) = \begin{cases} e^{i\theta-x}, & x \geq i\theta, \\ 0, & x < i\theta, \end{cases} \quad i = 1, \dots, n.$$

Equivalently, using an indicator function,

$$f_{X_i}(x | \theta) = e^{i\theta-x} \mathbf{1}\{x \geq i\theta\}.$$

Joint density

By independence, the joint density of $\mathbf{X} = (X_1, \dots, X_n)$ is

$$\begin{aligned} f_{\mathbf{X}}(\mathbf{x} | \theta) &= \prod_{i=1}^n f_{X_i}(x_i | \theta) \\ &= \prod_{i=1}^n (e^{i\theta-x_i} \mathbf{1}\{x_i \geq i\theta\}) \\ &= \left(\prod_{i=1}^n e^{i\theta-x_i} \right) \left(\prod_{i=1}^n \mathbf{1}\{x_i \geq i\theta\} \right) \\ &= \exp \left(\theta \sum_{i=1}^n i - \sum_{i=1}^n x_i \right) \mathbf{1}\{x_i \geq i\theta \text{ for all } i\}. \end{aligned}$$

Now

$$\{x_i \geq i\theta \text{ for all } i\} \iff \left\{ \frac{x_i}{i} \geq \theta \text{ for all } i \right\} \iff \left\{ \min_{1 \leq i \leq n} \frac{x_i}{i} \geq \theta \right\}.$$

Define the statistic

$$T(\mathbf{x}) = \min_{1 \leq i \leq n} \frac{x_i}{i}.$$

Then the indicator becomes $\mathbf{1}\{\theta \leq T(\mathbf{x})\}$. Hence we can rewrite

$$f_{\mathbf{X}}(\mathbf{x} \mid \theta) = \exp\left(\theta \sum_{i=1}^n i\right) \mathbf{1}\{\theta \leq T(\mathbf{x})\} \exp\left(-\sum_{i=1}^n x_i\right).$$

Apply the Factorization Theorem

Let

$$g_{\theta}(T(\mathbf{x})) = \exp\left(\theta \sum_{i=1}^n i\right) \mathbf{1}\{\theta \leq T(\mathbf{x})\}, \quad h(\mathbf{x}) = \exp\left(-\sum_{i=1}^n x_i\right).$$

Then

$$f_{\mathbf{X}}(\mathbf{x} \mid \theta) = g_{\theta}(T(\mathbf{x})) h(\mathbf{x}),$$

where g_{θ} depends on the sample $\mathbf{x}$ only through $T(\mathbf{x})$, and h does not involve θ .

By the Neyman–Fisher factorization theorem, $T = \min_{1 \leq i \leq n} (X_i/i)$ is a sufficient statistic for θ .

6.8 Let $X_1, X_2, \dots, X_n$ be a random sample from a population with location PDF $f(x - \theta)$. Show that the order statistics, $T(X_1, X_2, \dots, X_n) = (X_{(1)}, X_{(2)}, \dots, X_{(n)})$ are a sufficient statistic for θ and no further reduction is possible.

A population has a location PDF if there exists a fixed PDF f_0 such that for some unknown location parameter θ ,

$$f_X(x \mid \theta) = f_0(x - \theta), \quad \text{where } -\infty < \theta < \infty$$

Equivalently, if $Y \sim f_0$, then $X = \theta + Y$.

Location family: sufficiency and minimal sufficiency of the order statistics

Let $X_1, \dots, X_n$ be i.i.d. with density

$$f(x \mid \theta) = f_0(x - \theta), \quad -\infty < \theta < \infty,$$

where f_0 is a fixed PDF. Let $X_{(1)} \leq \dots \leq X_{(n)}$ be the order statistics and

$$T(\mathbf{X}) = (X_{(1)}, \dots, X_{(n)}).$$

1. Sufficiency of the order statistics

The joint density of $\mathbf{X} = (X_1, \dots, X_n)$ is

$$f(\mathbf{x} \mid \theta) = \prod_{i=1}^n f_0(x_i - \theta).$$

Let $\mathbf{x}_{()} = (x_{(1)}, \dots, x_{(n)})$ denote the sorted version of $\mathbf{x}$. Because multiplication is commutative,

$$\prod_{i=1}^n f_0(x_i - \theta) = \prod_{i=1}^n f_0(x_{(i)} - \theta).$$

Also, to map the ordered vector $\mathbf{x}_{()}$ back to $\mathbf{x}$ we only need to know which permutation produced $\mathbf{x}$; define

$$h(\mathbf{x}) = \mathbf{1}\{x_1, \dots, x_n \text{ are a permutation of } x_{(1)}, \dots, x_{(n)}\}.$$

Note: $h(\mathbf{x})$ always equals 1 since every valid set $(x_1, x_2, \dots, x_n)$ is a permutation of the order statistics. Basically, once you know the order statistics, any permutation of them has the same likelihood.

Then $h(\mathbf{x})$ does not involve θ , and we can write

$$f(\mathbf{x} | \theta) = g_\theta(T(\mathbf{x})) h(\mathbf{x}), \quad g_\theta(\mathbf{t}) = \prod_{i=1}^n f_0(t_i - \theta).$$

By the Neyman–Fisher factorization theorem, $T(\mathbf{X}) = (X_{(1)}, \dots, X_{(n)})$ is sufficient for θ .

Because the likelihood in a location model is **symmetric** in the data, permuting the observations does not change its value. So all the θ -information lives in the sorted sample; the ordering labels carry no θ -information.

2. No further reduction is possible (minimal sufficiency)

Use the likelihood-ratio characterization: a statistic T is minimal sufficient iff for any $\mathbf{x}, \mathbf{y}$,

$$\frac{f(\mathbf{x} | \theta)}{f(\mathbf{y} | \theta)} \text{ is independent of } \theta \iff T(\mathbf{x}) = T(\mathbf{y}).$$

Now,

$$\frac{f(\mathbf{x} | \theta)}{f(\mathbf{y} | \theta)} = \prod_{i=1}^n \frac{f_0(x_i - \theta)}{f_0(y_i - \theta)}.$$

If $\mathbf{y}$ is a permutation of $\mathbf{x}$, then the multisets $\{y_1, \dots, y_n\}$ and $\{x_1, \dots, x_n\}$ are identical, so the product above equals 1 for all θ . Hence the ratio is independent of θ .

Conversely, assume $\mathbf{y}$ is *not* a permutation of $\mathbf{x}$. Then the multisets $\{x_1, \dots, x_n\}$ and $\{y_1, \dots, y_n\}$ differ, so there exists a value a that appears with different multiplicity in the two samples. For a location family with nondegenerate f_0 , shifting θ changes the collection $\{x_i - \theta\}$ relative to $\{y_i - \theta\}$, and therefore the product $\prod_{i=1}^n f_0(x_i - \theta)/f_0(y_i - \theta)$ cannot remain constant in θ unless the two multisets coincide. Thus the ratio is independent of θ only when $\mathbf{y}$ is a permutation of $\mathbf{x}$.

Therefore,

$$\frac{f(\mathbf{x} | \theta)}{f(\mathbf{y} | \theta)} \text{ is independent of } \theta \iff (x_{(1)}, \dots, x_{(n)}) = (y_{(1)}, \dots, y_{(n)}),$$

which is exactly $T(\mathbf{x}) = T(\mathbf{y})$.

Hence $T(\mathbf{X}) = (X_{(1)}, \dots, X_{(n)})$ is minimal sufficient, so no further reduction is possible.

$(X_{(1)}, \dots, X_{(n)}) \text{ is sufficient and minimal sufficient for } \theta.$

6.9b For the following distribution, let $X_1, \dots, X_n$ be a random sample. Find a minimal sufficient statistic for θ .

$$f(x|\theta) = e^{-(x-\theta)}, \text{ where } \theta < x < \infty, -\infty < \theta < \infty$$

Minimal sufficient statistic for θ

Let $X_1, \dots, X_n$ be i.i.d. with density

$$f(x | \theta) = e^{-(x-\theta)} \mathbf{1}\{x > \theta\}, \quad -\infty < \theta < \infty.$$

Equivalently,

$$f(x | \theta) = e^{\theta-x} \mathbf{1}\{x > \theta\}.$$

Step 1: Write the joint density and factorize

The joint density of $\mathbf{X} = (X_1, \dots, X_n)$ is

$$f(\mathbf{x} | \theta) = \prod_{i=1}^n e^{-(x_i-\theta)} \mathbf{1}\{x_i > \theta\} = \exp\left(-\sum_{i=1}^n (x_i - \theta)\right) \prod_{i=1}^n \mathbf{1}\{x_i > \theta\} = \exp\left(n\theta - \sum_{i=1}^n x_i\right) \mathbf{1}\{\theta < \min(x_1, \dots, x_n)\}.$$

Let

$$X_{(1)} = \min(X_1, \dots, X_n).$$

Then

$$f(\mathbf{x} | \theta) = \underbrace{\exp(n\theta) \mathbf{1}\{\theta < X_{(1)}\}}_{g_\theta(X_{(1)})} \underbrace{\exp\left(-\sum_{i=1}^n x_i\right)}_{h(\mathbf{x})}.$$

By the Neyman–Fisher factorization theorem, $X_{(1)}$ is sufficient for θ .

Step 2: Show $X_{(1)}$ is minimal sufficient

Use the likelihood ratio characterization: a statistic T is minimal sufficient iff for any $\mathbf{x}, \mathbf{y}$,

$$\frac{f(\mathbf{x} | \theta)}{f(\mathbf{y} | \theta)} \text{ is independent of } \theta \iff T(\mathbf{x}) = T(\mathbf{y}).$$

Compute the ratio using the joint density above:

$$\frac{f(\mathbf{x} | \theta)}{f(\mathbf{y} | \theta)} = \frac{\exp(n\theta - \sum_{i=1}^n x_i) \mathbf{1}\{\theta < x_{(1)}\}}{\exp(n\theta - \sum_{i=1}^n y_i) \mathbf{1}\{\theta < y_{(1)}\}} = \exp\left(-\sum_{i=1}^n x_i + \sum_{i=1}^n y_i\right) \frac{\mathbf{1}\{\theta < x_{(1)}\}}{\mathbf{1}\{\theta < y_{(1)}\}},$$

where $x_{(1)} = \min_i x_i$ and $y_{(1)} = \min_i y_i$.

The exponential factor does not involve θ . Thus the ratio is independent of θ if and only if the indicator quotient is independent of θ . But if $x_{(1)} \neq y_{(1)}$, say $x_{(1)} < y_{(1)}$, then:

$$\theta \in (x_{(1)}, y_{(1)}) \implies \mathbf{1}\{\theta < x_{(1)}\} = 0 \text{ and } \mathbf{1}\{\theta < y_{(1)}\} = 1,$$

so the ratio equals 0 for those θ ; while for $\theta < x_{(1)}$ both indicators are 1 and the ratio equals $\exp(-\sum x_i + \sum y_i)$. Hence the ratio depends on θ . Therefore, for the ratio to be independent of θ we must have $x_{(1)} = y_{(1)}$.

Conversely, if $x_{(1)} = y_{(1)}$, then the indicator terms are identical for all θ and the ratio reduces to

$$\frac{f(\mathbf{x} | \theta)}{f(\mathbf{y} | \theta)} = \exp \left(- \sum_{i=1}^n x_i + \sum_{i=1}^n y_i \right),$$

which is independent of θ .

Thus,

$$\frac{f(\mathbf{x} | \theta)}{f(\mathbf{y} | \theta)} \text{ is independent of } \theta \iff x_{(1)} = y_{(1)}.$$

By the likelihood ratio criterion, $X_{(1)}$ is minimal sufficient.

$T(\mathbf{X}) = X_{(1)} = \min(X_1, \dots, X_n) \text{ is a minimal sufficient statistic for } \theta.$

Extra Problem:

Suppose that X_1 and $X_2 \sim \text{Bernoulli}(p)$. Let $T(\mathbf{X}) = (X_1 + X_2)$. Show that:

- T is sufficient for p .
- T is also minimal sufficient.

Bernoulli sample: sufficiency and minimal sufficiency of $T = X_1 + X_2$

Let $X_1, X_2 \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p)$ with $0 < p < 1$, and define $T(\mathbf{X}) = X_1 + X_2$.

1. T is sufficient for p

The joint pmf of (X_1, X_2) is

$$f(x_1, x_2 | p) = p^{x_1} (1-p)^{1-x_1} p^{x_2} (1-p)^{1-x_2} = p^{x_1+x_2} (1-p)^{2-(x_1+x_2)}.$$

Since $T = x_1 + x_2$, this can be written as

$$f(x_1, x_2 | p) = g_p(T) \quad \text{where} \quad g_p(t) = p^t (1-p)^{2-t}.$$

And,

$$h(x_1, x_2) = 1$$

By the Neyman–Fisher factorization theorem, $T = X_1 + X_2$ is sufficient for p .

2. T is minimal sufficient

A statistic T is minimal sufficient iff for all $\mathbf{x}, \mathbf{y}$,

$$\frac{f(\mathbf{x} | p)}{f(\mathbf{y} | p)} \text{ is independent of } p \iff T(\mathbf{x}) = T(\mathbf{y}).$$

Let $\mathbf{x} = (x_1, x_2)$ and $\mathbf{y} = (y_1, y_2)$. Then

$$\frac{f(\mathbf{x} \mid p)}{f(\mathbf{y} \mid p)} = \frac{p^{x_1+x_2}(1-p)^{2-(x_1+x_2)}}{p^{y_1+y_2}(1-p)^{2-(y_1+y_2)}} = \left(\frac{p}{1-p}\right)^{(x_1+x_2)-(y_1+y_2)}.$$

This expression is independent of p if and only if

$$x_1 + x_2 = y_1 + y_2 \iff T(\mathbf{x}) = T(\mathbf{y}).$$

Therefore $T = X_1 + X_2$ is minimal sufficient.

$T = X_1 + X_2$ is sufficient and minimal sufficient for p .
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